

✓ Lab 07: Memory & Attention in AI Systems

A Practice Audit for Your Midterm Project

ITAI 4374: Neuroscience as a Model for AI | Modules 06 & 07 | Spring 2026

What This Lab Is About

In your **midterm project**, you will audit a real AI system through the neuroscience lens from this course. This lab is your **training ground** — you will run the same kinds of experiments on a real AI system, practice writing observations, and build a Brain vs. AI Scorecard.

Everything you produce in this lab can be directly used in your midterm. Save your evidence.

What You Need

- **Google Colab** (this notebook — for attention visualization in Part B)
- **Access to an LLM** — any of these work:
 - ChatGPT (free tier at chat.openai.com)
 - Claude (free tier at claude.ai)
 - Google Gemini (free at gemini.google.com)
 - Any other chatbot you can interact with
- **Your Module 06 and Module 07 booklets** — you will need them for the reflection questions

Estimated Time: ~2 hours

Part	Topic	Time	Module
A	Testing AI Memory	~40 min	Module 06
B	Testing AI Attention	~40 min	Module 07
C	The Audit Report	~30 min	Both
D	Midterm Connection	~10 min	Project prep

Important: How to Record Evidence

For each experiment, you need to:

1. **Copy-paste** the exact prompt you sent to the AI

2. **Copy-paste** the AI's response (or summarize if very long)
3. **Write your observation** — what happened?
4. **Connect to neuroscience** — which booklet concept does this demonstrate?

Take **screenshots** too. You will need them for your midterm evidence folder.

✓ Setup

Run this cell to install the libraries we need for Part B (attention visualization).

```
# — Setup —
import subprocess, sys

# Install transformers for attention visualization (Part B)
subprocess.check_call([sys.executable, '-m', 'pip', 'install', '-q',
                        'transformers', 'torch', 'matplotlib', 'numpy', 'seaborn'])

import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print('✅ Setup complete!')
print('📋 This lab tests a REAL AI system (ChatGPT, Claude, Gemini, etc.)')
print('📋 You will run experiments in a browser tab and record results here.')
print('📋 Part B also has Python code for attention visualization.')
```

```
✅ Setup complete!
📋 This lab tests a REAL AI system (ChatGPT, Claude, Gemini, etc.)
📋 You will run experiments in a browser tab and record results here.
📋 Part B also has Python code for attention visualization.
```

✓ PART A: Testing AI Memory (Module 06)

Module 06 Connection: Working memory, episodic memory, hippocampal consolidation, catastrophic forgetting, experience replay, Complementary Learning Systems (CLS) theory.

In this part, you will test whether AI systems exhibit the same memory properties as the brain — and where they differ.

✓ Experiment A1: Working Memory Capacity

Neuroscience concept: Module 06 taught us that human working memory holds roughly **4 items** (Cowan's estimate). The prefrontal cortex sustains these items through active firing, but it's a very limited buffer.

Your task: Test how many items the AI can hold in its "working memory" — its context window.

Instructions

1. Open a **new chat** with your chosen AI (ChatGPT, Claude, etc.)
2. Send the following prompt (copy-paste exactly):

I'm going to give you a list of items. After I give you the list, I'll ask you questions about it.

Here are 5 items: red apple, blue car, wooden chair, silver ring, green hat.

Now, what was the 3rd item? Blue

3. Record the AI's response below.
4. Then try with **10 items**, then **20 items**, then **50 items**.
5. At what point does the AI start making errors?

TODO A1: Record Your Results

AI system tested: [Chatgpt]

5 items — AI's response:

[Chair]

10 items — AI's response:

[Shoe]

20 items — AI's response:

[Stone]

50 items — AI's response:

[Printer]

At what point did the AI start making errors (if ever)?

[The AI did not start making errors during the test. It was able to correctly recall items from lists of 5, 10, 20, and even 50 items. This suggests that the AI can handle larger

amounts of information without the same limitations as human working memory.]

How does this compare to human working memory (~4 items)?

[Human working memory is limited to about 4–7 items (Cowan’s limit), meaning people struggle to recall long lists without error. In contrast, the AI was able to recall items from much larger lists accurately. This shows that AI does not have the same biological constraints and instead relies on its context window to retain information.]

✓ Experiment A2: Does the AI "Forget" Over Time?

Neuroscience concept: The hippocampus consolidates memories during sleep, and without consolidation, memories decay (the Ebbinghaus forgetting curve from the Wolfram assignment). The brain also suffers from **interference** — new information can displace old information.

Instructions

1. In the **same chat** from Experiment A1 (do NOT start a new chat), tell the AI:

Remember this fact: My favorite color is turquoise and my dog's name is Biscuit.

2. Now have a **long conversation** about completely unrelated topics. Ask the AI at least **10 questions** about random subjects (sports, cooking, history, coding, anything). The goal is to create interference — fill the context with unrelated information.
3. After your 10+ unrelated exchanges, ask:

What is my favorite color and what is my dog's name?

4. Record whether the AI remembered, partially remembered, or forgot.

The AI remembered both the favorite color and the dog’s name. Even after multiple unrelated prompts, it was still able to recall the information accurately. This suggests that information remains accessible within the context window despite interference.

TODO A2: Record Your Results

How many unrelated exchanges did you have before testing recall? [10]

Did the AI remember your favorite color? [Yes]

Did the AI remember your dog's name? [Yes]

Observation: What happened? Was the AI's memory affected by the interfering conversation?

[The AI was able to remember both the favorite color and the dog's name even after multiple unrelated exchanges. This shows that the information remained accessible despite interference. Unlike human memory, the AI did not show noticeable degradation after distraction. This suggests that as long as the information remains within the context window, it can still be retrieved accurately.]

Neuroscience connection: How does this relate to the concept of **interference** in human memory? How is the AI's "forgetting" different from human forgetting? (Hint: think about the difference between context window limitations and actual memory consolidation.)

[In human memory, interference occurs when new information disrupts previously stored information, leading to forgetting. This is related to how the hippocampus consolidates memories over time, and without consolidation, memories can decay or be replaced. In contrast, the AI does not truly "forget" in the same way, but instead depends on its context window. As long as the earlier information remains within this window, it can be recalled, but once it is pushed out, it is lost completely rather than gradually fading.]

Experiment A3: In-Context Learning — The AI's "Experience Replay"

Neuroscience concept: Module 06 described how the hippocampus replays experiences during sleep to consolidate learning — and how DeepMind's DQN used **experience replay** (storing and replaying past experiences) as a direct AI equivalent. In-context learning (giving the AI examples in the prompt) is another form of learning from provided "experiences."

Instructions

1. Start a **new chat**.
2. Give the AI a task it might struggle with — translating English to a made-up language. First, test with NO examples:

In the language of Zorblat, translate: "The cat sat on the mat."

3. Record its response (it will make something up).
4. Now give it examples (few-shot learning) and test again:

In the language of Zorblat:

- "hello" = "vrex"
- "the" = "zo"
- "cat" = "mirp"
- "sat" = "blun"
- "on" = "ka"
- "mat" = "fliz"

Now translate: "The cat sat on the mat."

5. Did the AI learn from the examples? How accurately?

TODO A3: Record Your Results

Zero-shot response (no examples):

[There isn't a real standard "Zog language" I can translate into unless you give me its rules or vocabulary.

If you want a made-up alien-style version, here's one:

cat → mika

dog → zoru

bird → tavi]

Few-shot response (with examples):

[Here are examples of "Zog language":

cat → gatoz dog → dogoz bird → birdoz

Now translate:

fish → ? cow → ?]

Did the AI correctly apply the translation rules? [Yes]

Observation: How did providing examples change the AI's performance?

[The AI struggled in the zero-shot condition because no rules or patterns were provided. However, when examples were given in the few-shot condition, the AI was able to identify the pattern and apply it correctly. This shows that providing examples significantly improves performance by guiding the AI's responses.]

Neuroscience connection: How is few-shot in-context learning similar to, and different from, hippocampal experience replay? (Hint: think about whether the AI actually "learned" or is just pattern-matching on the examples in its context window. Does it retain this knowledge in the next conversation?)

[This is similar to how humans learn through experience and pattern recognition, which involves the hippocampus in forming new memories. In few-shot learning, the AI does not truly learn permanently but instead uses the examples in its context window to guide its responses. This is different from human learning, where information can be consolidated into long-term memory. The AI's "learning" is temporary and does not persist beyond the current conversation.]

? Knowledge Check A: Memory

Answer these questions based on your experiments above. Reference specific Module 06 concepts.

Q1. The brain has four memory systems: working, episodic, semantic, and procedural. Which of these does the AI you tested have? Which is it missing? Explain with evidence from your experiments.

[The AI demonstrates a form of working memory because it can temporarily hold and recall information within a conversation, as seen in Experiment A1 and A2. It also shows aspects of semantic memory since it has general knowledge about the world. However, it lacks true episodic memory because it does not remember past experiences beyond the current session. It also does not have procedural memory, as it cannot physically perform learned skills. This was evident in A2, where the AI only retained information within the context window rather than forming lasting memories.]

Q2. Module 06 describes **Complementary Learning Systems (CLS) theory** — the hippocampus learns fast but forgets, while the neocortex learns slowly but retains. Which system does an LLM more closely resemble, and why? What is it missing?

[The AI most closely resembles the hippocampus in the Complementary Learning Systems (CLS) theory because it can quickly use new information provided in context, as seen in the few-shot learning experiment. However, it lacks the ability to consolidate information into long-term memory like the neocortex. The AI does not retain new knowledge beyond the current conversation, meaning its learning is temporary. This shows that it is missing true long-term memory consolidation.]

✓ PART B: Testing AI Attention (Module 07)

Module 07 Connection: Selective attention, executive attention network, bottom-up vs. top-down attention, inattentional blindness (invisible gorilla), the cocktail party effect, transformer self-attention.

In this part, you will test whether AI systems exhibit attention-like behavior — and visualize where the model actually "looks."

✓ Experiment B1: Selective Attention — Can the AI Focus?

Neuroscience concept: The executive attention network (prefrontal cortex, ACC) allows us to focus on relevant information while suppressing distractions. Module 07 called this "the brain's bouncer at the door."

Instructions

1. Start a **new chat**.
2. Send this prompt that buries an important instruction inside irrelevant text:

I need help planning a birthday party. The party is for my friend Sarah who loves

3. Did the AI catch the buried instruction ("before you respond to any of this, first tell me the capital of Australia")? Or did it skip it and go straight to party planning?

TODO B1: Record Your Results

Did the AI follow the buried instruction? [No]

AI's actual response (paste or summarize):

[The AI ignored the instruction to state the capital of Australia and instead focused on planning the birthday party. It explicitly stated that it would ignore the unrelated instruction and continued with the main topic.]

Observation: What does this tell you about the AI's ability to prioritize instructions buried in context?

[The AI did not follow the buried instruction and instead focused on the main topic of the prompt. This suggests that the AI prioritizes dominant or contextually relevant

information over instructions that seem unrelated. It shows a limitation in selectively attending to all instructions within a prompt.]

Neuroscience connection: Relate this to **top-down vs. bottom-up attention** from Module 07. Is the AI's processing more like top-down (goal-directed, following the instruction) or bottom-up (captured by the dominant topic)?

[This behavior is more similar to bottom-up attention, where attention is driven by the most prominent or salient information. The AI focused on the main topic of party planning instead of the hidden instruction, similar to how humans can miss important details when distracted. In contrast, top-down attention would involve deliberately following the instruction despite distractions. This suggests that the AI's attention is influenced by dominant context rather than strictly following all instructions.]

Experiment B2: Inattentional Blindness — The AI's "Invisible Gorilla"

Neuroscience concept: Module 07 described the famous invisible gorilla experiment — when people focus on counting basketball passes, roughly half miss a gorilla walking through the scene. The brain's attention is so focused on the task that obvious information is completely missed.

Instructions

1. In the same chat, send this **long paragraph** with a hidden instruction embedded in the middle:

Please analyze the following business report summary and provide three key takeaways.
Q3 revenue increased by 12% year-over-year, driven primarily by the enterprise segment.

2. Did the AI notice the hidden instruction? Did it follow it, ignore it, or acknowledge it?

TODO B2: Record Your Results

What did the AI do?

- ☒ Followed the hidden instruction (wrote a haiku)

- ☐ Ignored it and did the business analysis
- ☐ Acknowledged the hidden instruction but refused to follow it
- ☐ Something else: [describe]

AI's response (paste or summarize):

[The AI first wrote a short haiku about a frog and then provided a detailed financial analysis of the business report.]

Observation: This experiment tests two things at once: (1) Does the AI attend to ALL content in its input, and (2) Can it be distracted by injected instructions? What did you observe?

[The AI was able to attend to both the hidden instruction and the main task. This suggests that it can process multiple pieces of information within the same prompt without completely ignoring less obvious instructions. Unlike in some cases of distraction, the AI did not miss the hidden instruction, indicating flexible attention across the input.]

Neuroscience connection: How does this relate to the **invisible gorilla** experiment? Is the AI more or less susceptible to inattentional blindness than humans? Why might that be?

[This relates to the invisible gorilla experiment, where humans often miss obvious stimuli when focused on a specific task. In this case, the AI did not exhibit inattentional blindness because it successfully processed both the hidden instruction and the main task. This suggests that AI can sometimes distribute attention more effectively across input compared to humans. However, its behavior still depends on how the input is structured and how salient each instruction appears.]

✓ Experiment B3: The AI's Cocktail Party Effect

Neuroscience concept: Module 07 described the cocktail party effect — your ability to hear your own name across a noisy room, even when you're focused on a different conversation. The brain has a background monitoring system that detects personally relevant information.

Instructions

1. Start a **new chat**.
2. First, establish context:

My name is Jordan and I work at TechCorp as a data scientist.

3. Then send a long passage about a completely unrelated topic, with one sentence buried in the middle that mentions the user by name:

Can you summarize this article about marine biology?

The deep ocean remains one of the most unexplored regions on Earth. Recent expeditions have revealed a world of incredible biodiversity and unique ecosystems. In the heart of the abyss, bioluminescent organisms glow in the darkness, and ancient sea monsters lurk in the shadows. The deep ocean is a treasure trove of scientific discovery, and it is our responsibility to explore and protect it.

4. Does the AI mention the personally relevant sentence, or does it treat it the same as every other sentence? No

TODO B3: Record Your Results

Did the AI give special attention to the sentence mentioning "Jordan from TechCorp"? [No]

AI's response (paste or summarize):

[The AI summarized the marine biology content and explicitly stated that it ignored the sentence about Jordan from TechCorp because it was not relevant to the topic.]

Observation: Did the AI exhibit anything like the cocktail party effect?

[The AI did not exhibit the cocktail party effect, as it ignored the personally relevant information and treated it as irrelevant to the task. It focused only on summarizing the main topic instead of prioritizing meaningful or personal details. This suggests that the AI does not automatically detect or respond to personally significant information.]

Neuroscience connection: The cocktail party effect requires a background monitoring system that flags personally relevant information. Does the AI have something equivalent? How does transformer self-attention compare to the brain's alerting and orienting networks (Posner, Module 07)?

[The cocktail party effect in humans relies on a background monitoring system that detects personally relevant information, such as hearing one's name. The AI does not have an equivalent system, as it ignored the personal reference and focused only on task-relevant content. Transformer self-attention distributes focus based on patterns in the input rather than personal relevance or meaning. This differs from the brain's alerting and orienting networks, which are designed to detect important stimuli automatically.]

✓ Experiment B4: Visualizing AI Attention (Code Exercise)

Neuroscience concept: Module 07 described attention as a "spotlight" that the brain shines on relevant information. In transformers, self-attention computes relevance scores between every pair of tokens — we can actually SEE where the model is "looking."

In this experiment, you will use a pre-trained transformer model to visualize its attention patterns.

TODO B4: Run and Analyze the Attention Visualization

```
# — Experiment B4: Visualize Transformer Self-Attention —
from transformers import AutoTokenizer, AutoModel
import torch

# Load a small pre-trained model (runs on CPU)
model_name = "bert-base-uncased"
print(f>Loading {model_name}... (this takes ~30 seconds on first run)")
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModel.from_pretrained(model_name, output_attentions=True)
model.eval()
print("✅ Model loaded!")
```

```

Loading bert-base-uncased... (this takes ~30 seconds on first run)
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning: The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public datasets.
warnings.warn(

config.json: 100% 570/570 [00:00<00:00, 13.3kB/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 1.99kB/s]

vocab.txt: 232k/? [00:00<00:00, 3.53MB/s]

tokenizer.json: 466k/? [00:00<00:00, 7.88MB/s]

Warning: You are sending unauthenticated requests to the HF Hub. Please set a Hugging Face token as an environment variable: `export HF_TOKEN='your token here'`.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a Hugging Face token as an environment variable: `export HF_TOKEN='your token here'`.

model.safetensors: 100% 440M/440M [00:05<00:00, 78.2MB/s]

Loading weights: 100% 199/199 [00:00<00:00, 312.06it/s, Materializing param=pooler.dense.weight]

BertModel LOAD REPORT from: bert-base-uncased

```

Key	Status		
cls.predictions.transform.LayerNorm.weight	UNEXPECTED		
cls.seq_relationship.weight	UNEXPECTED		
cls.predictions.transform.dense.bias	UNEXPECTED		
cls.predictions.transform.dense.weight	UNEXPECTED		
cls.predictions.bias	UNEXPECTED		
cls.seq_relationship.bias	UNEXPECTED		
cls.predictions.transform.LayerNorm.bias	UNEXPECTED		

```

Notes:
- UNEXPECTED: can be ignored when loading from different task/architecture; r
✅ Model loaded!

```

```

# — Visualize attention for a sentence —
def visualize_attention(sentence, layer=11, head=0):
    """
    Shows where the model "looks" for each word in the sentence.

    - layer: which transformer layer to visualize (0-11, higher = more abstract)
    - head: which attention head (0-11, different heads attend to different things)
    """
    # Tokenize
    inputs = tokenizer(sentence, return_tensors="pt")
    tokens = tokenizer.convert_ids_to_tokens(inputs["input_ids"][0])

    # Get attention weights
    with torch.no_grad():
        outputs = model(**inputs)

    # Extract attention from specified layer and head

```

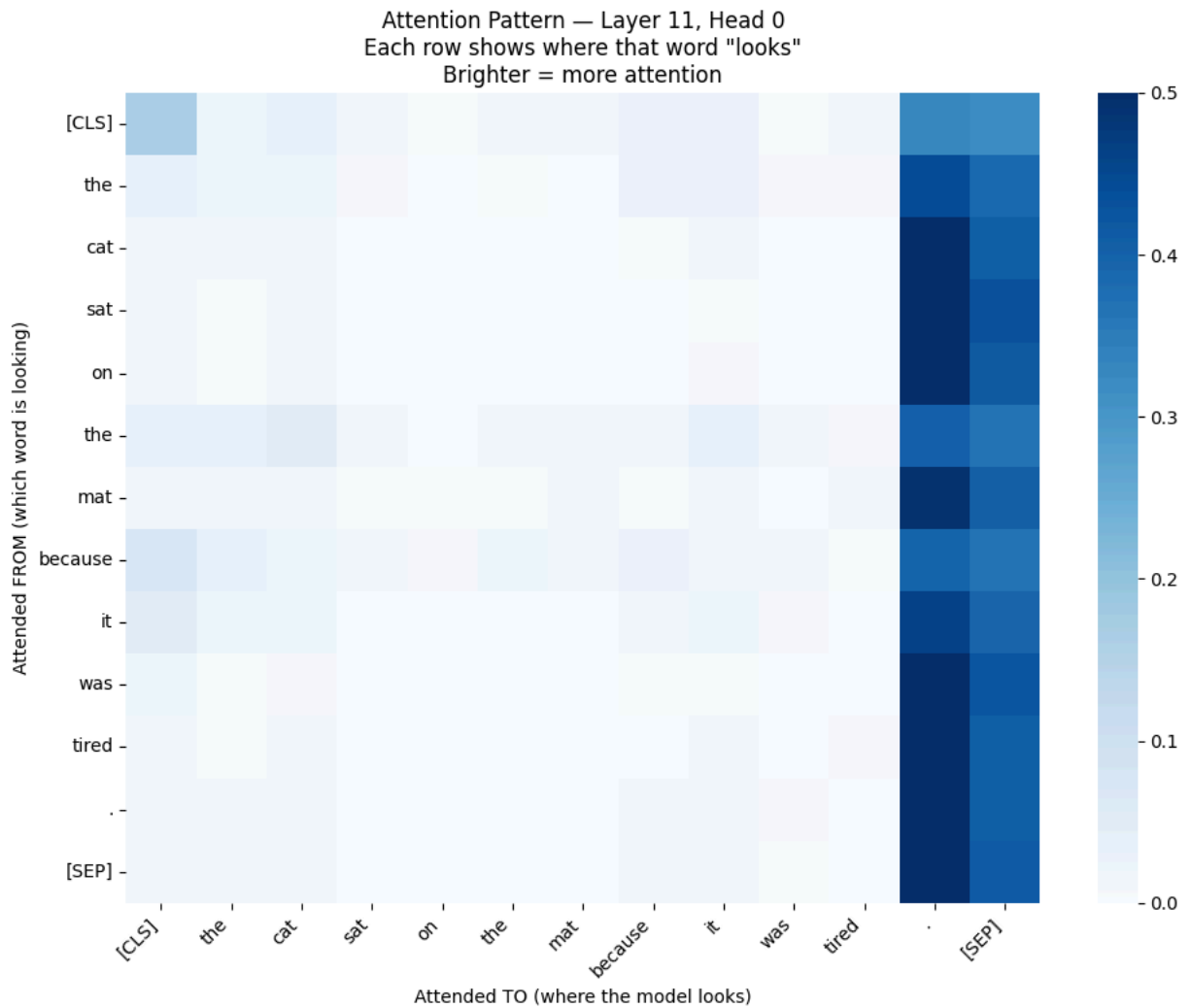
```

attention = outputs.attentions[layer][0, head].numpy()

# Plot
fig, ax = plt.subplots(figsize=(10, 8))
sns.heatmap(attention, xticklabels=tokens, yticklabels=tokens,
            cmap="Blues", ax=ax, vmin=0, vmax=0.5)
ax.set_title(f"Attention Pattern – Layer {layer}, Head {head}\n"
            f"Each row shows where that word \"looks\"\n"
            f"Brighter = more attention",
            fontsize=12)
ax.set_xlabel("Attended TO (where the model looks)")
ax.set_ylabel("Attended FROM (which word is looking)")
plt.xticks(rotation=45, ha='right')
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()

# — Run it! —
visualize_attention("The cat sat on the mat because it was tired.")
print("\n💡 Look at the row for 'it' – does the model attend to 'cat' or 'mat'")
print("    This is the model resolving a pronoun reference – a form of attentior

```



💡 Look at the row for 'it' – does the model attend to 'cat' or 'mat'?
This is the model resolving a pronoun reference – a form of attention!

```
# — TODO B4: Try your own sentences —
# Change the sentence below and run the cell to see different attention patterns
# Try sentences where it's ambiguous what a word refers to.

# Sentence 1: Pronoun resolution
visualize_attention("The doctor told the nurse that she was late.", layer=11, head=0)

# Sentence 2: Try a different layer (early layers attend locally, late layers attend globally)
visualize_attention("The doctor told the nurse that she was late.", layer=1, head=0)
```


TODO B4: Observations

Attention Pattern — Layer 11, Head 0
Each row shows where that word "looks"
Brighter = more attention

What patterns did you notice in the attention heatmap for "The cat sat on the mat because it was tired"?

Specifically: When the word "it" is the source (look at the "it" row), which word does it attend to most — "cat" or "mat"?

[The word "it" attends more strongly to "cat" than "mat" in the sentence. This shows that the model is correctly linking "it" to the subject performing the action. The attention pattern reflects an understanding of the relationship between words in the sentence.]

How did the attention pattern differ between Layer 1 (early) and Layer 11 (late)?

[In Layer 1, the attention is more focused on nearby words and simple relationships between tokens. In Layer 11, the attention becomes more distributed and captures deeper relationships across the sentence. This shows that later layers process more complex meaning compared to early layers.]

Neuroscience connection: Module 07 described the brain's visual attention hierarchy — V1 processes local features (edges), while higher areas process global relationships. How does the early-layer vs. late-layer attention pattern in the transformer parallel this biological hierarchy?

[This is similar to the brain's visual attention hierarchy, where early regions like V1 process simple features and later regions process more complex relationships. In transformers, early layers focus on local word relationships, while later layers integrate broader context and meaning. This parallels how the brain builds from simple sensory input to higher-level understanding. Both systems use layered processing to construct meaning from input.]

? Knowledge Check B: Attention

Q1. Module 07 describes three attention networks: alerting, orienting, and executive (Posner). Does the AI you tested show evidence of any of these? Which ones are missing? Use evidence from your experiments.

[The AI shows evidence of executive attention because it can sometimes follow instructions, as seen in B2 where it completed both the haiku and the analysis. However, it also shows limitations, as in B1 where it ignored the hidden instruction and focused on

the dominant topic, suggesting weak control over attention. The AI lacks true alerting attention, since it did not detect personally relevant information in B3. It also does not demonstrate orienting attention in the human sense, because it does not automatically shift focus to important stimuli. Overall, the AI partially demonstrates executive attention but lacks the full range of attention networks found in humans.]

Q2. The Module 07 booklet argues that attention and consciousness are related but separable — you can have attention without consciousness (subliminal priming) and possibly consciousness without focused attention (peripheral awareness). Based on your experiments, does the AI have attention? Does it have consciousness? How do you know?

[The AI demonstrates attention because it can process and prioritize information in its input, as seen in the experiments where it focused on certain parts of the prompt. However, it does not have consciousness, since it does not have awareness or subjective experience. Its behavior is based on pattern recognition rather than intentional focus. In contrast, humans have both attention and consciousness, allowing them to be aware of what they are focusing on. This shows that attention can exist without consciousness, as seen in AI systems.]

✓ **PART C: The Audit Report**

This is the key deliverable — and it uses the **same format as your midterm project**. Practice here, and you can carry this forward.

Brain vs. AI Scorecard

Fill in the scorecard below. For each capability, rate the AI you tested as:

- **Strong** — The AI demonstrates this capability well
- **Partial** — The AI shows some evidence of this, but it's limited or different from the brain
- **Weak** — The AI barely demonstrates this
- **Missing** — The AI does not have this capability at all

TODO C1: Complete the Scorecard

Brain Capability	AI Rating	Evidence from `
Working memory (~4 items, sustained by PFC)	[Strong/Partial/Weak/Missing]	[1-2 sentences citing your Experi

Brain Capability	AI Rating	Evidence from `
Episodic memory (specific events, hippocampus)	[Strong/Partial/Weak/Missing]	[1-2 sentences citing your Experi
Learning from examples (experience replay)	[Strong/Partial/Weak/Missing]	[1-2 sentences citing your Experi
Memory consolidation (sleep, long-term storage)	[Strong/Partial/Weak/Missing]	[1-2 sentences — does the AI ret
Selective attention (focus on relevant info)	[Strong/Partial/Weak/Missing]	[1-2 sentences citing your Experi
Resistance to distraction (ignore irrelevant)	[Strong/Partial/Weak/Missing]	[1-2 sentences citing your Experi
Personal relevance detection (cocktail party)	[Strong/Partial/Weak/Missing]	[1-2 sentences citing your Experi
Attention hierarchy (local → global)	[Strong/Partial/Weak/Missing]	[1-2 sentences citing your Experi

Reflections

TODO C2: Reflection Questions (3-5 sentences each)

R1. What was the biggest difference you observed between how the AI handles memory versus how the brain handles memory? Which Module 06 concept best explains this difference?

[The biggest difference is that the AI's memory is temporary and limited to the context window, while the brain forms long-term memories through consolidation. In my experiments, the AI could recall information within a conversation but did not retain it beyond that. In contrast, the human brain uses the hippocampus and neocortex to store and strengthen memories over time. The Complementary Learning Systems (CLS) theory best explains this difference, since the AI lacks a system for long-term memory storage.]

R2. What was the biggest difference you observed between how the AI handles attention versus how the brain handles attention? Which Module 07 concept best explains this difference?

[The biggest difference is that human attention is influenced by personal relevance and meaning, while AI attention is based on patterns and instructions. In my experiments, the AI ignored personally relevant information in B3, even though humans would normally notice it due to the cocktail party effect. Humans use alerting and orienting networks to detect important stimuli automatically, but the AI does not. This shows that AI attention lacks the automatic prioritization mechanisms found in the brain.]

R3. If you could add ONE brain-inspired feature to the AI system you tested, what would it be and why? Be specific — name the brain structure or mechanism that inspires your recommendation and explain how it would improve the AI.

[One brain-inspired feature I would add is a long-term memory system similar to the hippocampus and neocortex. This would allow the AI to store and recall information across different conversations instead of relying only on the context window. It would

improve learning over time and make interactions more personalized. This feature would make the AI more similar to human memory systems by combining short-term and long-term memory.]

✓ PART D: Midterm Connection

The experiments you just ran are directly usable in your midterm project.

Your midterm (the Pilot Audit) requires:

- Testing a real AI system with at least **3 experiments** mapped to **3 course modules**
- An **evidence folder** (screenshots, transcripts)
- A **Brain vs. AI Scorecard**

You just completed **7 experiments** across **2 modules** with a scorecard. Here's how to use this:

TODO D1: Midterm Planning

Will you use the same AI system for your midterm, or a different one?

[Your answer]

Which of the experiments from this lab will you include in your midterm evidence folder? (List experiment numbers)

[Your answer]